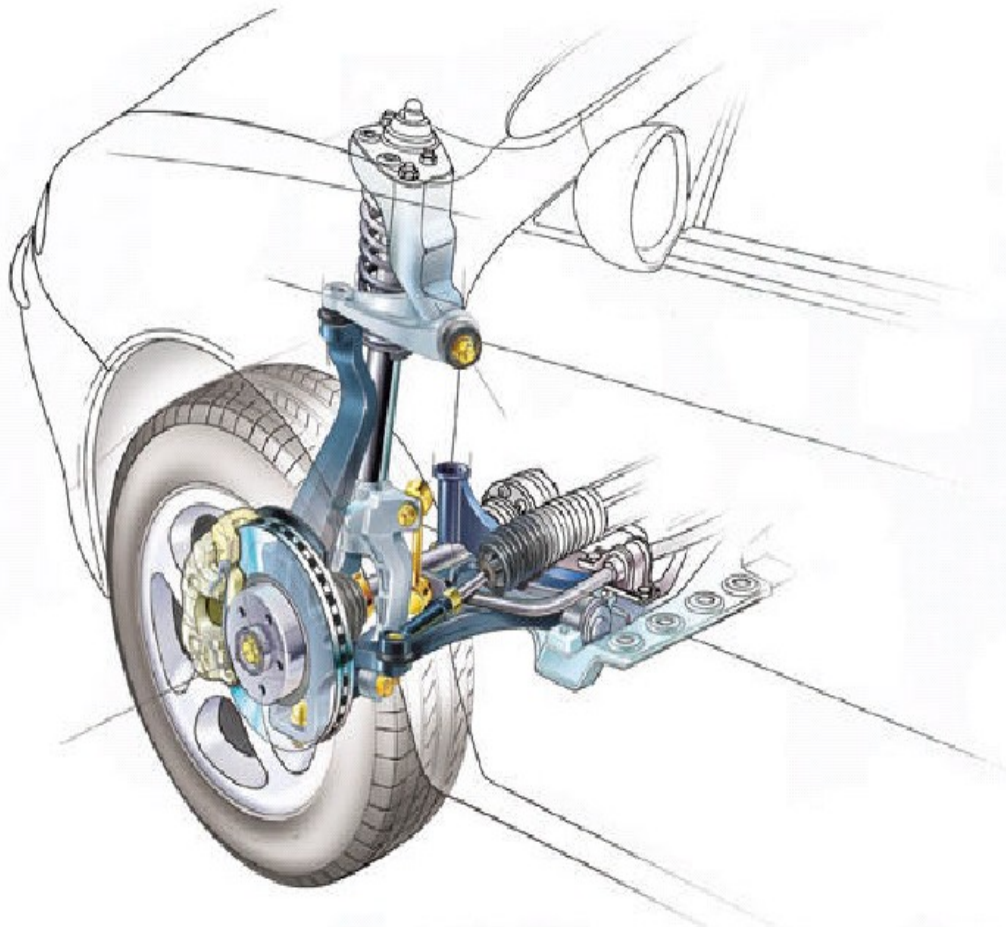


SUSPENSION SYSTEM IN AUTOMOBILES

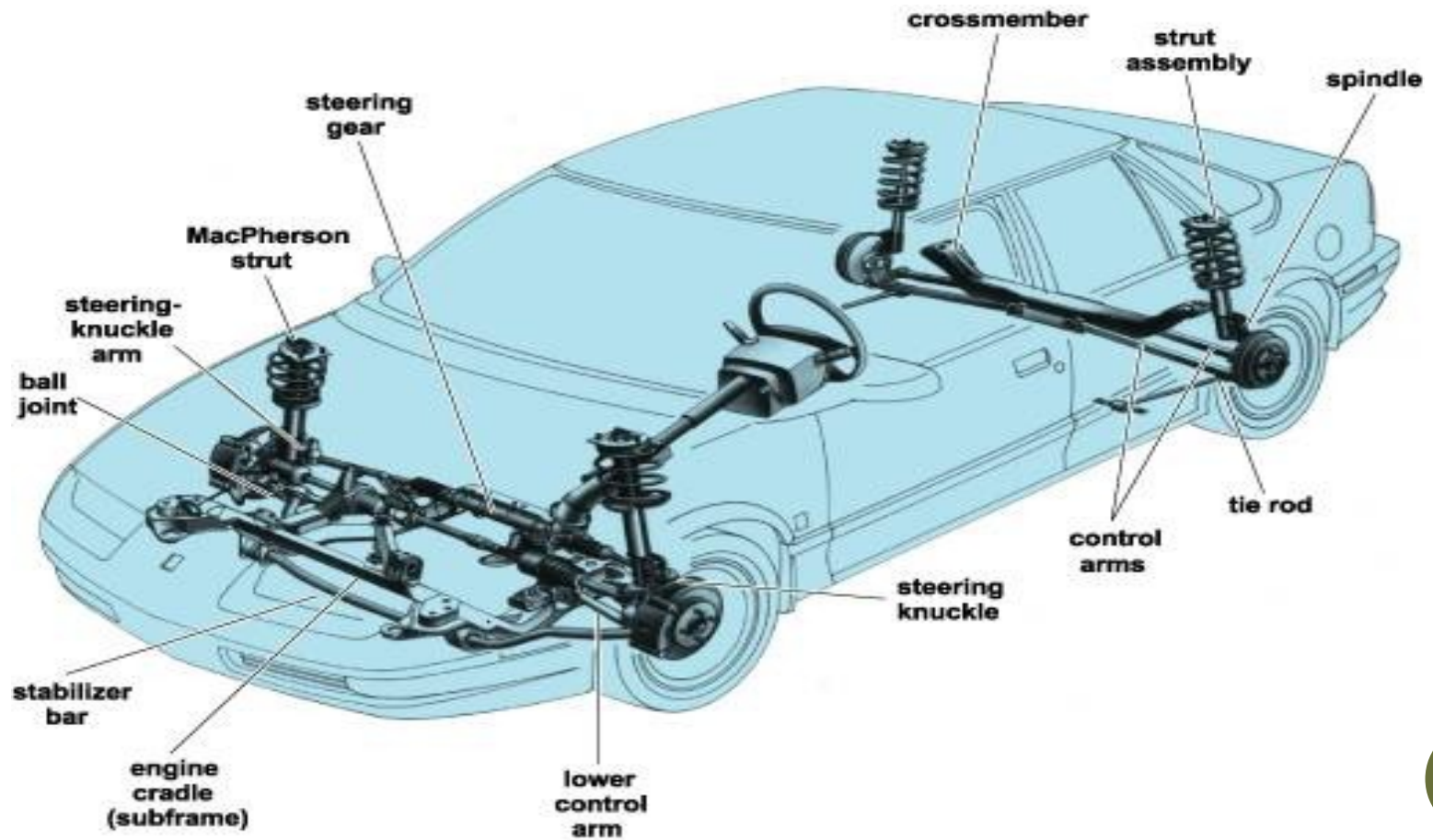


WHAT IS SUSPENSION SYSTEM

- **Suspension** is the term given to the system of springs, shock absorbers and linkages that connects a vehicle to its wheels
- Serve a dual purpose – contributing to the car's handling and braking.
- Protects the vehicle itself and any cargo or luggage from damage and wear



SUSPENSION SYSTEM – LAY OUT



DIFFERENT SUSPENSION SYSTEMS

- Conventional suspension system
- Independent suspension system
- Air suspension system
- Hydro elastic suspension system



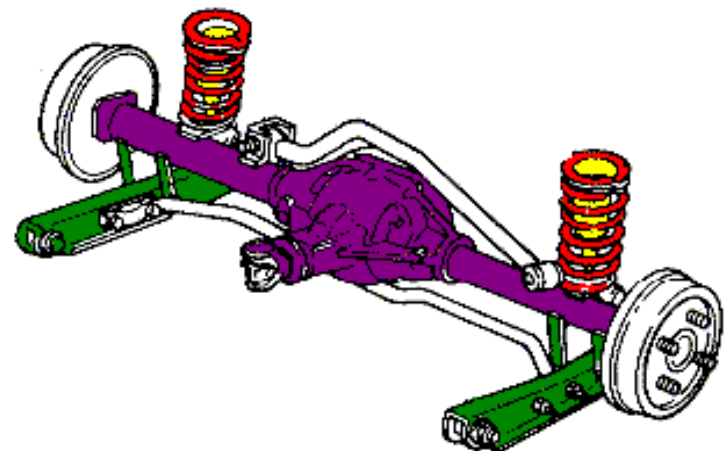
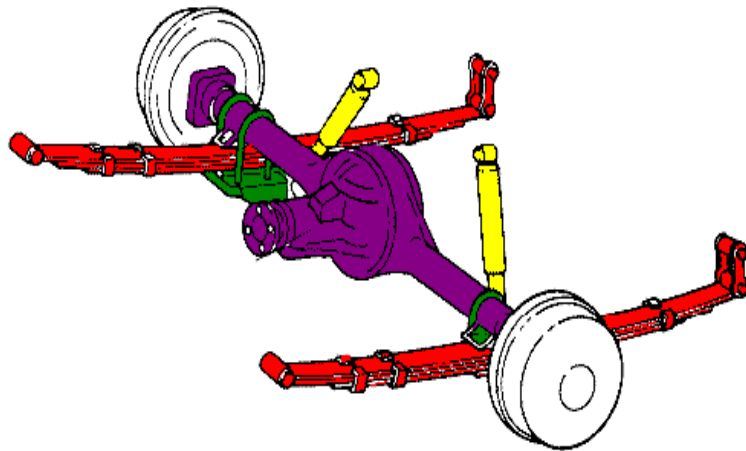
CONVENTIONAL SUSPENSION SYSTEM

- Two wheels are mounted on either side of the rigid axle
- When one wheel encounters the bump, both the wheel do not execute parallel up and down motion
- So it gives rise to gyroscopic effect and wheel wobble
- Rear driving wheels mounted on live axle suspended by laminated leaf springs and shock absorbers





D97/6991



INDEPENDENT SUSPENSION SYSTEM

- Both the front and the rear wheel are utilized
- Design incorporated in the front wheels
- One wheel goes down ,the other wheel does not have much effect
- Basic classification of the design
 1. **MacPherson Strut**
 2. **Double Wishbone**
 3. **Multi link**



MACPHERSON STRUT

- the most widely used front suspension system in cars
- comprises of a strut-type spring and shock absorber combo, which pivots on a ball joint on the single, lower arm.
- The steering gear is either connected directly to the lower shock absorber housing, or to an arm from the front or back of the spindle (in this case
- When you steer, it physically twists the strut and shock absorber housing (and consequently the spring) to turn the wheel



DOUBLE WISHBONE SUSPENSION

Type of *double-A* or *double wishbone* suspension

Wheel spindles are supported by an upper and lower 'A' shaped arm.

The lower arm carries most of the load.

If you look head-on at this type of system, parallelogram system that allows the spindles to travel vertically up and down.

This side-to-side motion is known as scrub



- Type of *double-A arm* suspension although the lower arm in these systems can sometimes be replaced with a single solid arm (as in my picture).
- The spring/shock combo is moved from between the arms to above the upper arm.
- This transfers the load-bearing capability of the suspension almost entirely to the upper arm and the spring mounts.
- The lower arm in this instance becomes a control arm.



MULTI-LINK SUSPENSION

- It's currently being used in the Audi A8 and A4 amongst other cars.
- The basic principle of it is the same, but instead of solid upper and lower wishbones, each 'arm' of the wishbone is a separate item.
- These are joined at the top and bottom of the spindle thus forming the wishbone shape.
- The super-weird thing about this is that as the spindle turns for steering, it alters the geometry of the suspension by torquing all four suspension arms.
- Spring is separate from the shock absorber.



ADVANTAGES (INDEPENDENT FRONT)

- Bigger deflection of front wheels, no reaction on steering
- Greater distance for resisting rolling action
- Front axle (small-stub), improves road holding tendency of tyres.
- Minimum vibrations

DISADVANTAGES

- Better shock absorber required.
- Expensive
- Tyre wear increases due to transmission of torque.



ADVANTAGES (INDEPENDENT REAR SUSPENSION)

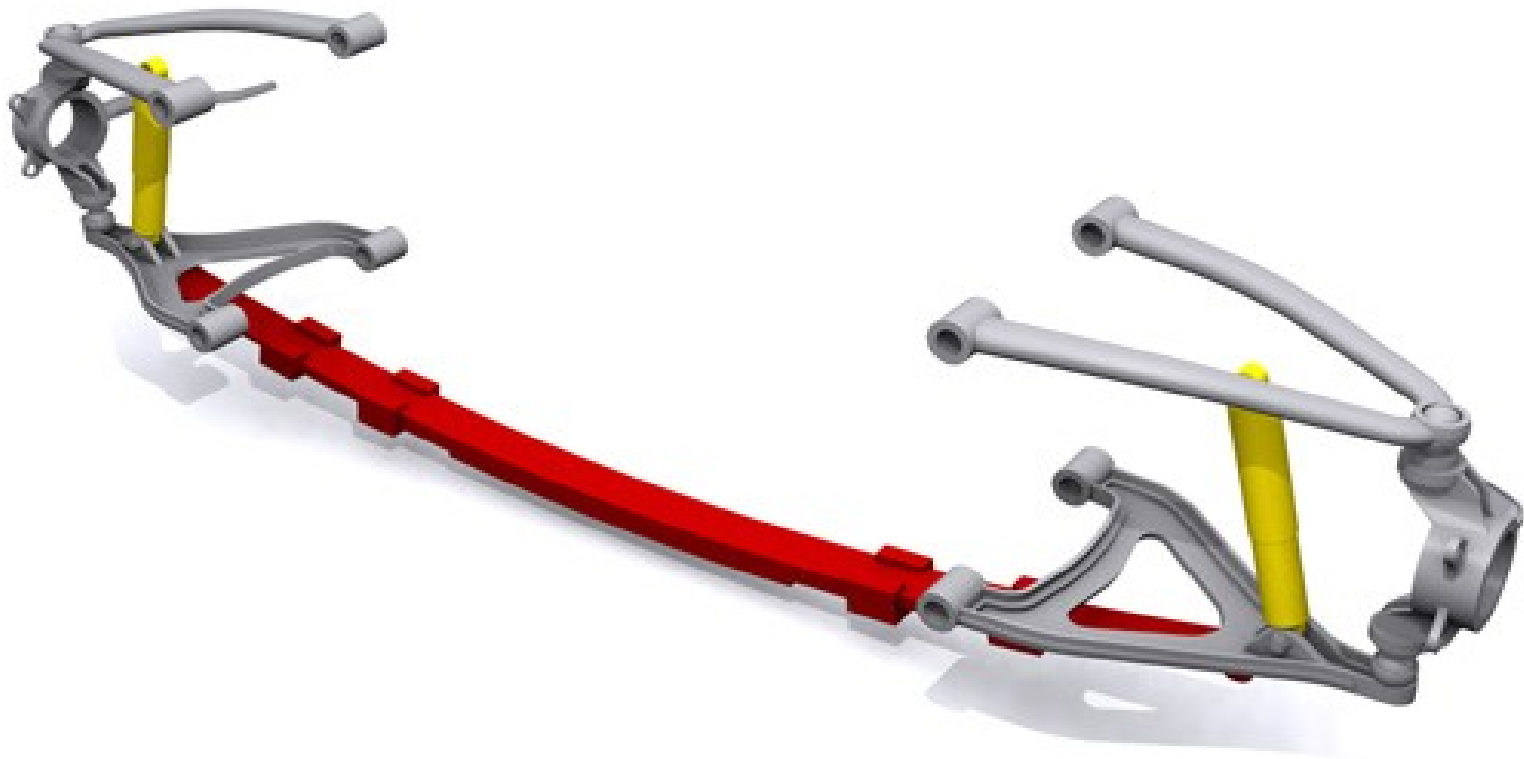
- Lesser unsprung weight – improves ride , reduces tyre wear.
- Increased passenger space
- Rear wheels remain stable ..

DISADVANTAGES

- Increased cost
- Complicated design
- Steering action is not proper



TRAVERSE SUSPENSION SYSTEM



- Normally find on the rear suspension
- Combines independent double wishbone suspension with a leaf spring.
- It involves one leaf spring mounted *across* the vehicle, connected at each end to the lower wishbone.
- The centre of the spring is connected to the front subframe in the middle of the car.
- There are still two shock absorbers, mounted one to each side on the lower wishbones.

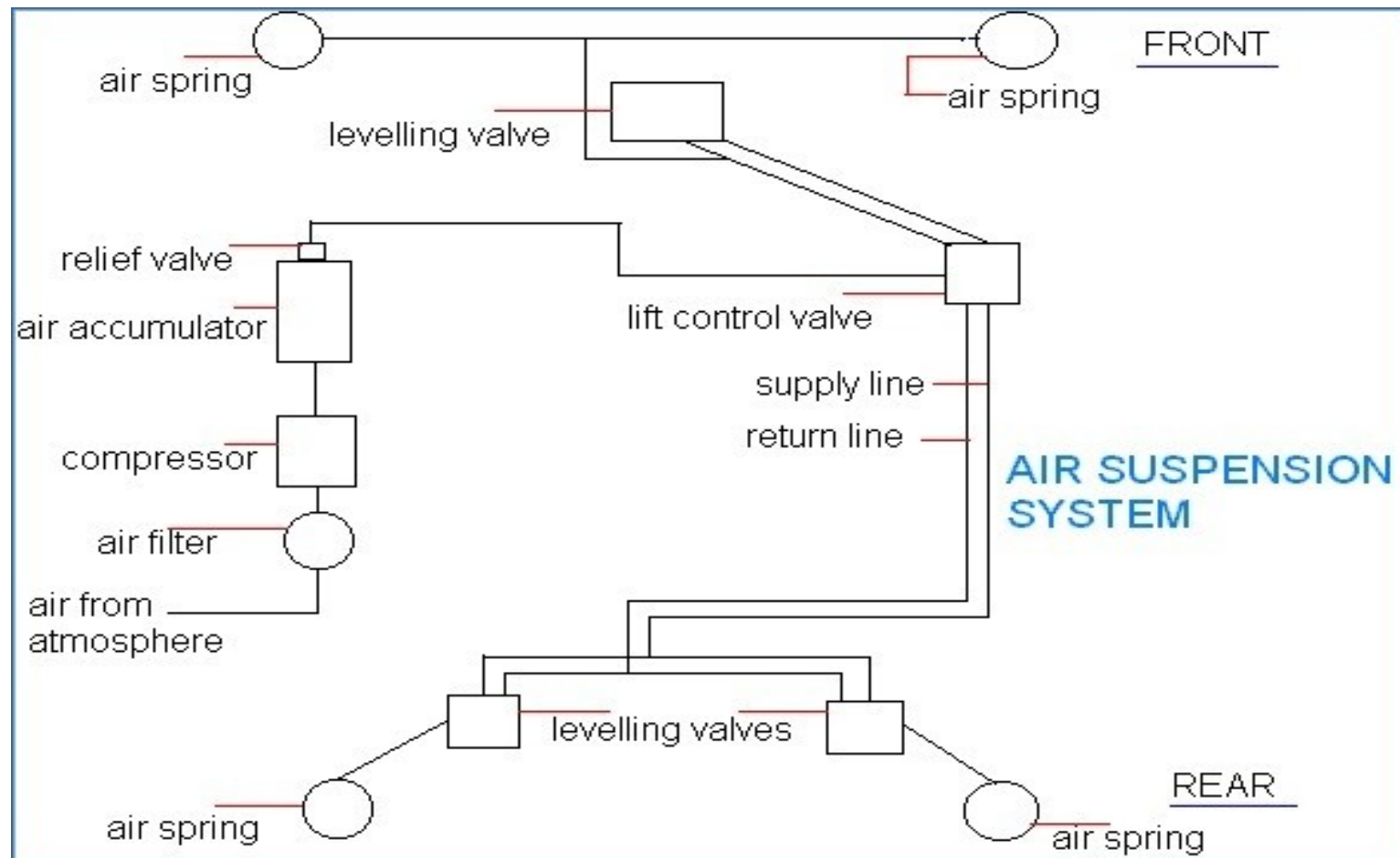


AIR SUSPENSION

- Comprises of compressor , suppling air to air tank
- Pressure maintained – 5.6 to 7 kg/sq.m
- Air bags – on each wheel
- As load applied , air bags compressed actuating the levelling valve .
- Air from the tank fills the compressrd air bag & hence raise the level of the frame.
- Air from air bag gets released as load on chassis decreases .



AIR SUSPENSION LAYOUT



ADVANTAGES OF AIR SUSPENSION

- These maintain a constant frequency of vibration whether the vehicle is laden or unladen.
- Constant frame height is maintained.
- It helps to reduce the load while the vehicle in motion i.e. the dynamic loading as the spring rate variation between laden and unladen weight is much less.
- It gives smooth and comfort ride of the vehicle.
- The stiffness of the system increases with the increase of the deflection.



COMMON AIR SUSPENSION PROBLEMS

❑ Air bag or air strut failure

- due to old age, or moisture within the air system that damages them from the inside.

❑ Compressor failure

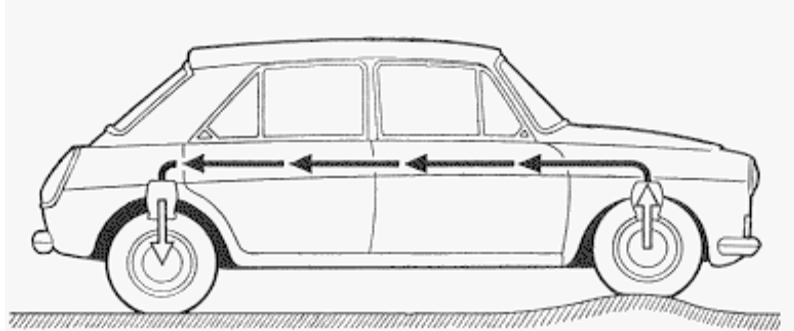
- Primarily due to leaking air springs or air struts
- Compressor burnout may also be caused by moisture from within the air system coming into contact with its electronic parts.

❑ Dryer failure

- which functions to remove moisture from the air system eventually becomes saturated and unable to perform that function

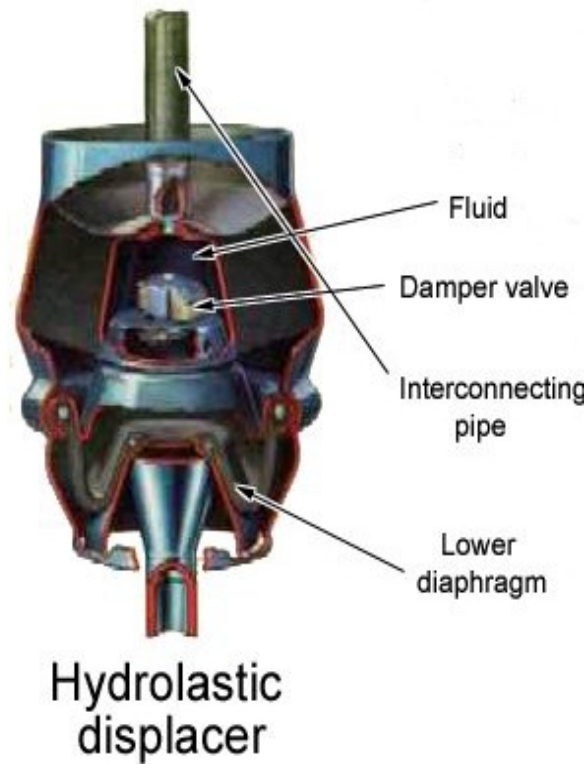


HYDROLASTIC SUSPENSION



- a system where the front and rear suspension systems were connected together in order to better level the car when driving.
- The front and rear suspension units have Hydrolastic displacers, one per side.
- These are interconnected by a small bore pipe. Each displacer incorporates a rubber spring
- Damping of the system is achieved by rubber valves.





- when the front wheel encounter bumps ,the piston moves upwards pressurising the fluid to enter into the rear unit.
- Hydroelastic was eventually refined into Hydragas suspension.....

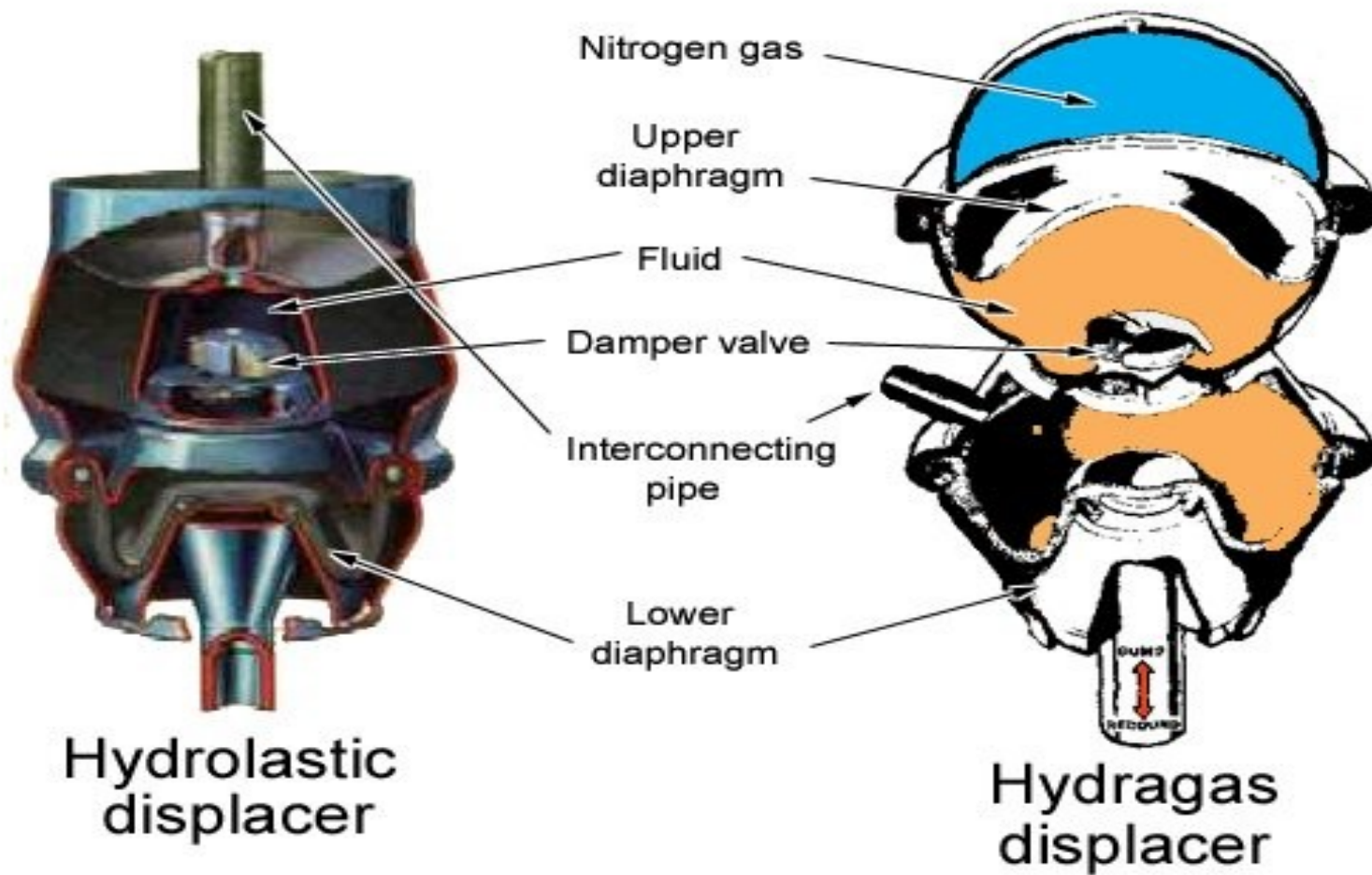


HYDROGAS SUSPENSION

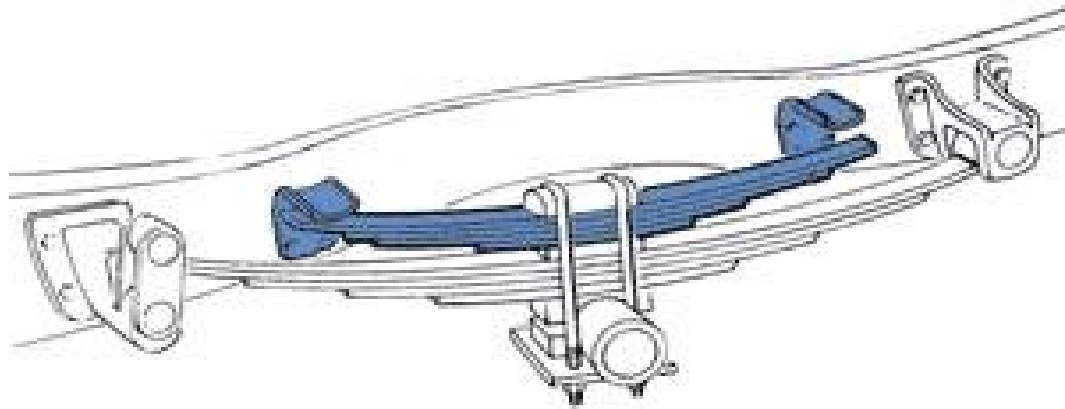
- Known as hydro-pneumatic suspension.
- The difference is in the displacer unit itself.
- In the older systems, fluid was used in the displacer units with a rubber spring cushion built-in.
- With Hydragas, the rubber spring is removed completely.
- The fluid still exists but above the fluid there is now a separating membrane or diaphragm, and above that is a cylinder or sphere which is charged with nitrogen gas.
- The nitrogen section is what has become the spring and damping unit whilst the fluid is still free to run from the front to the rear units and back.



DIFFERENCE ...



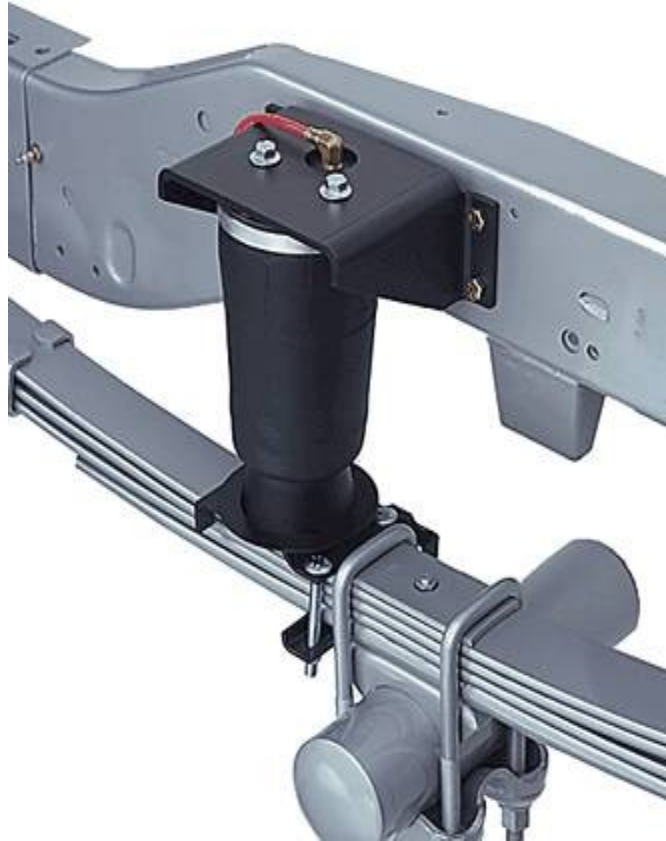
HELPER SPRING



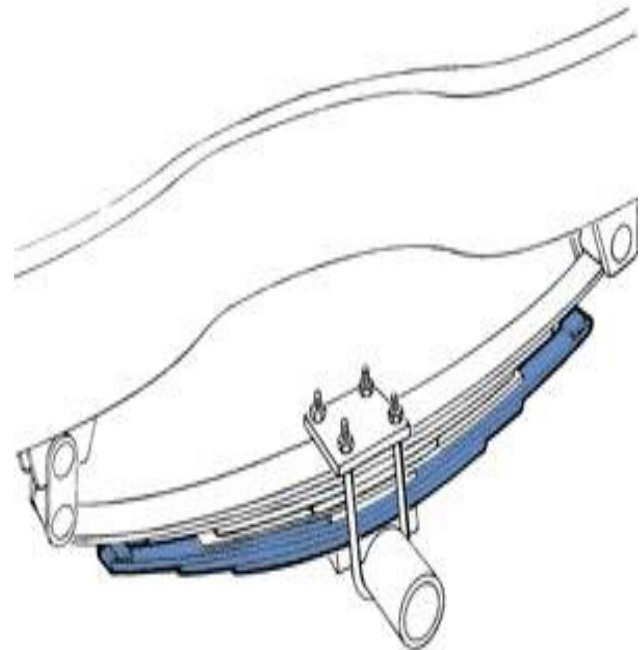
- ◉ DIRECTLY MOUNTED ON MAIN SPRINGS
- ◉ TAKE CARE OF LARGE VARIATION IN SPRING LOAD
- ◉ DURING LIGHT LOADS ,ONLY MAIN SPIRNG IS ACTIVE , AS LOAD INCREASE TO A PARTICULAR FIXED VALUE , BOTH THE SPRINGS ARE ACTIVE



AIR ASSISTED HELPER SPRING



LEAF HELPER SPRING



THANK YOU !!!

